

The pointwise Stiefel-transform conjecture is already answered

Literature status for arXiv:1009.2184

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Status: literature already answered. Rubin’s direct sequel, arXiv:1103.1161, explicitly answers the pointwise questions from Part I and proves the analytic continuation, smoothness, composition, and Funk-inversion assertions of Conjecture 6.8.

Source question

B. Rubin, *Funk, Cosine, and Sine Transforms on Stiefel and Grassmann manifolds*, arXiv:1009.2184, Conjecture 6.8 on PDF page 38. For a smooth right $O(m)$ -invariant function f on $V_{n,m}$ (with $2m \leq n$ in the composition/Funk setting), the conjecture asks whether the normalized cosine family $\mathcal{M}^\alpha f$

- (a) extends pointwise meromorphically with possible polar set $\mathbb{N} = \{1, 2, \dots\}$;
- (b) is smooth and right $O(m)$ -invariant away from $\mathbb{N}$;
- (c) satisfies, pointwise,

$$\mathcal{M}^{2m-\alpha-n} \mathcal{M}^\alpha f = f \quad (\alpha, 2m - \alpha - n \notin \mathbb{N});$$

- (d) gives the pointwise Funk inversion $\mathcal{M}^{2m-n} F_m f = \varkappa_m f$.

Exact later answer

B. Rubin, *Funk, Cosine, and Sine Transforms on Stiefel and Grassmann manifolds, II*, arXiv:1103.1161. Its abstract and introduction on PDF pages 1–2 say that the Part-I identities, previously proved only as distributions, are now obtained pointwise, and that the paper answers the corresponding questions from Part I.

The clauses match as follows. Theorem 3.1(iv), on PDF pages 7–8, proves that the cosine transform divided by $\Gamma_m(\alpha/2)$ is entire in α and smooth in the Stiefel variable. After multiplication by the remaining normalizing factor, the paragraph before Theorem 6.4 on PDF page 13 states that $\mathcal{M}^\alpha f$ is defined for every $\alpha \notin \mathbb{N}$ and is a smooth function. This proves (a)–(b) in the intended possible-pole sense. Theorem 6.4 on PDF pages 13–14 is exactly the pointwise identity in (c). Finally, Theorem 6.1 on PDF pages 12–13 gives the pointwise inversion formula for $F_{m,k}$; taking $k = m$ is exactly clause (d), with the same constant.

Identification and scope

The sequel cites Part I as its reference [34], uses the same operators and normalization, and explicitly presents its results as pointwise upgrades of Part I. This is therefore an author-identified answer, not a new deduction. The words “the polar set” cannot literally assert that every positive integer is an actual pole for every f (already $f = 0$ is a counterexample); both papers use $\mathbb{N}$ as the universal possible/excluded polar set. Part II also poses new problems, including explicit Wallach-set measures and higher-rank Sobolev range descriptions; those are separate and are not claimed resolved here.